MILITARY SPECIFICATION

INDICATORS; AIRSPEED, PITOT-STATIC PRESSURE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for all types of 2 .750 inch round dial pitot-static airspeed indicators.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

DD-G-451	Glass, Float or Plate, Sheet, Figured (Flat, For Glazing, Mirrors And Other Uses)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-C-320	Chromium Plating (Electrodeposited)
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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MIL-M-3171	Magnesium Alloy, Processes For Corrosion Protection Of
MIL-N-3336	Nut, Self-Locking, Instrument Mounting
MIL-C-5541	Chemical Films For Aluminum And Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
DOD-D-1000	Drawings, Engineering And Associated Lists
MIL-C-14806	Coating, Reflection Reducing, For Instrument Cover Glass And Lighting Wedges
MIL-P-26514	Polyurethane Foam, Rigid Or Flexible, For Packaging
MIL-C-83488	Coating, Aluminum, ION Vapor Deposited

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-2073-1	DOD Materiel Procedures For Development And Application Of Packaging Requirements
MS28105	Window, Dial-Aircraft Instrument Cover, Glass
MS33558	Numeals and Letters, Aircraft Instrument Dial, Standard Form Of

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

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(Nongovernment standards and other publications are noramlly available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the refernces cited (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Type of testing. The type of testing (qualification or preproduction) to be accomplished on indcators shall be as specified in the detail specification.

3.2 Data. Unless otherwise specified in the contract or order, no data are required by this specification or any of the documents referenced in section 2 herein (see 6.2).

3.3 Selection of Government documents. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts like screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractors drawing.

3.3.2 Standard parts. With the exception of 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lighest practicable weight and suitable for the purpose intended.

3.4.1 Critical materials. Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be of the least critical of those which are adequate for the purpose.

3.4.2 Metals. Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt fog, or atmospheric conditions to which the instrument may be subject when in storage or during normal service life.

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3.4.2.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means or protective coatings or hermetic sealing.

3.4.3 Nonmagnetic materials.Nonmagnetic material shall be used for all parts of the instruments, except where magnetic materials are essential.

3.4.4 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity.

3.4.5 Corrosion fumes. Materials used in the interior construction of the indicators shall not produce corrosive or deleterious fumes.

3.5 Design and construction. The design of the indicator shall be as specified in the detail specification. The design shall be reasonably simple to facilitate as as much as possible disassembly, repair or overhaul, serivce maintenance, and reassembly by those tools and items of maintenance equipment which are normally available as commercial standards without requiring special tools and fixtures. The indicator shall be so constucted that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service, without failure.

3.5.1 Case. The indicator case shall be in accordance with the detail specification. The case shall be made of lightweight, nonmagnetic metal, uniform in texture, having a smooth surface, covered with a durable black finish and shall be pressure tight. The indicator case shall be so designed that the pressure connection is centered and the static connection is off center.

3.5.1.1 Reinforecement. The inside of the case around each insert shall be strengthened by a reinforcement to avoid damage to the case when the connection to either insert is tightened during the installation of the indicator. If the inserts are constructed of aluminum alloy, they shall be anodically treated, except for the threads.

3.5.2 Dial plate. The dial plate shall be made of nonmagnetic metal. The dial shall be fastened securely into the case or upon the frame of the mechanism by at least two screws in such a manner that it will not loosen or turn when the indicator is vibrated.

3.5.2.1 Dial markings. Dial markings shall conform to the applicable figure of the detail specification. The style and proportions of the letters and numerals placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies. The painting material shall be of a quality sufficient to insure durability of the markings in service use.

3.5.3 Cover glass. The cover glass shall be flat and its quality shall be in accordance with type II, quality AA of DD-G-451. The glass shall not be fluorescent when exposed to ultraviolet light or rays in the 365-millimicron region. The dimensions shall be in accordance with MS28105-16.

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3.5.3.1 Cover glass mounting. The cover glass shall be properly sealed and means shall be provided both for its replacement and to prevent fogging or the accumulation of moisture on the inside surface under extremes of atmospheric conditions. The distance between the front surface of the mounting flange and the outside surface of the cover glass shall not exceed 0.062 inch. The distance between the dial plate and the inside surface of the cover glass shall not exceed 0.125 inch for indicators requiring one pointer or 0.187 inch for indicators requiring two pointers.

3.5.3.2 Anti-reflection coating. A coating in accordance with MIL-C-14806 shall be applied to each surface of the cover glass (and the lighting wedge if applicable).

3.6 Pointers. The pointers shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechansim, and shall be readily adjustable.

3.7 Application of pressure. The pitot pressure shall be applied to the inside of the diaphragm capsule and the static pressure to the interior of the case.

3.7.1 Capillary tubing. A sufficient length of approved capillary tubing having a minimium inside diameter of 0.020 inch, or some other acceptable restriction, shall be incorporated in the instrument case between the pitot connection in the rear of the case and the diaphragm capsule. The effect of this restriction shall be such that the indicator will pass the damping tests of 4.6.6 and as specified in the detail specification.

3.7.2 Standard pressure. The standard pressure used in calibrating airspeed indicators for use with pitot-static tubes shall be in accordance with the values specified in U.S. Standard Atmosphere, 1962.

3.7.3 Plug. Each static and pitot pressure connection shall be provided with a vented plug of material and a design acceptable to the contracting activity. The vent hole shall be 0.063 $\pm$ 0.016 inch in diameter.

3.8 Physical requirements.

3.8.1 Protective treatment. When material are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent conformance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.8.2 Magnesium alloy parts. Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment shall be used. The treatment shall be approved by the contracting activity.

3.8.3 Aluminum alloy parts. Where practicable, aluminum alloy part shall be covered with an anodic film conforming to MIL-A-8625 except that the dial, if of aluminum alloy, small holes, pipe threads, and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

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3.8.4 Iron and steel parts. Iron and steel parts shall be nickel or chromium plated in accordance with QQ-N-290 or QQ-C-320, respectively.

3.8.4.1 Steel parts. As an alternative to nickel or chromium plating, steel parts may be coated with ion vapor deposited aluminum, in accordance with MIL-C-83488, Type I or II as applicable and of a class that is applicable to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.8.5 Machine screws. Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter shall conform to MIL-S-7742.

3.8.6 Material and size. Materials and dimensions of markings will be as specified in the detail specification.

3.9 Performance. The indicator shall satisfy the performance requirements as specified in Section 4 and in the detail specification.

3.10 Identification of product.

3.10.1 Nameplate. A nameplate conforming to MIL-STD-130 shall be securely attached to the exterior of the case and shall be durably marked to withstand usage encountered in service.

3.10.2 Manufacturer's part number. The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.11 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of DOD-D-1000.

3.12 Use of military designations. Military designations shall not be applied to a product, except for qualification test samples, nor referred to in correspondence or sales matter, until notification has been received from the activity responsible for qualification.

3.13 Installation.

3.13.1 Installation instructions. When specified in the contract or order, the contractor shall furnish with each instrument one printed copy of instructions, with illustrations and diagrams if necessary, covering the installation of the instrument. This information shall incldue data for front and back mounting installation of the indicator. Previous to printing, two copies shall be furnished to the contracting activity for approval. Whenever possible, the instructions shall be arranged to require only one sheet of paper, either 8 .500 by 11 inches or 17 by 11 inches. (see 6.2)

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3.13.2 Mounting screws. The contractor shall furnish sufficient mounting screws of the size and type specified on the applicable standard for installing the indicator.

3.13.3 Removeable spring nuts. When required, a sufficient number of removable No. 6-32 NC-2 spring nuts fit the mounting lugs of the instrument, of a type conforming to MIL-N-3336, shall be furnished (see 6.2).

3.13.4 Envelope. An envelope furnished by the contractor containing installation instructions and mounting screws shall be packaged with each indicator, and shall be marked with the following information:

IMPORTANT
THIS ENVELOPE CONTAINS
INSTRUCTIONS AND
MOUNTING SCREWS

3.13.4.1 Envelope and spring nuts. If removeable spring nuts are furnished, an envelope furnished by the contractor containing installation instructions, mounting screws, and spring nuts shall be packaged with each indicator, and shall be marked with the following information:

IMPORTANT
THIS ENVELOPE CONTAINS INSTRUC-
TIONS, MOUNTING SCREWS AND
SPRING NUTS

3.14 Workmanship. The instrument, including all parts and accessories, shall be constucted and finished to produce an instrument free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.15 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as other specified in the contract or prucahse order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements; in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3)

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection samples. The qualification inspection samples shall consist of three samples representative of the production equipment. The samples shall be identified with the manufacturer's part number and any additional information required by the contracting activity.

4.4 Quality conformance inspections. The quality conformance inspections shall consist of individual tests and sampling plans and tests.

4.4.1 Individual tests. Each indicator shall be subjected to the following tests in the order specified:

(a) Examination of product--(4.6.1)
(b) Leakage----------(4.6.2)
(c) Scale error at room temperature----------(4.6.3)
(d) Friction error----------(4.6.4)
(e) Position error----------(4.6.5)
(f) Low temperature operation----------(4.6.3.1)
(g) High temperature operation----------(4.6.3.2)

4.4.2 Sampling plans and tests.

4.4.2.1 Sampling Plan A. The sampling plan A tests shall consist of the following tests, which shall be conducted in the order specified, on each sample:

(a) Individual test--------(4.4.1)
(b) Damping----------(4.6.6)
(c) Weight----------(4.6.7)
(d) Magnetic effect--------(4.6.8)
(e) Pointer zero position error----------(4.6.9)

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4.4.2.1.1 Plan A Samping. Samples shall consist of 3 indicators selected at random from each lot of 100 or less which have passed the individual tests (4.4.1). A lot shall consist of identical indicators with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at same time. Indicators which have been subjected to the sampling plan A tests shall not be delivered on contract until they have been reconditioned and resubmitted to all individual tests.

4.4.2.2 Sampling plan B tests. Sampling plan B tests shall consist of the following tests which shall be conducted in the order specified:

(a) Sampling plan A tests---(4.4.2.1)
(b) Vibration----------(4.6.12)
(c) Overpressure----------(4.6.13)
(d) Acceleration----------(4.6.14)
(e) Humidity----------(4.6.15)
(f) Salt fog----------(4.6.16)
(g) Seasoning----------(4.6.17)
(h) Fungus ----------(4.6.18)
(i) Aging----------(4.6.19)
(j) Mounting nuts----------(4.6.20)
(k) Mounting lugs----------(4.6.21)
(l) Low temperature exposure(4.6.10)
(m) High temperature exposure----------(4.6.11)

4.4.2.2.1 Plan B sampling. Unless othewise specified, the samples shall consist of 3 indicators selected at random from the first 15 indicators, produced on the contract within 10 days after manufacture. Samples shall be forwarded to the laboratory specified (see 6.2) identified as required by the contracting activity.

4.4.3 Rejection and retest. Failure of the indicator to pass any test specified shall be cause for rejection. The instructions on reworking, refurbishing, or resubmitting rejected items shall be as specified in the general provisions of the contract or order.

4.5 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with test conditions specified in 4.5.1 of this specification.

4.5.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches of mercury) and at room temperature (approximately 25°C (77°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.5.2 Tapping. Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

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4.5.3 Attitude. Unless otherwise specified, the indicator shall be tested in normal operating position.

4.5.4 Vibration. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3,000 cycles per minute and shall subject the indicator to such vibration that a point on the case will describe, in a plan inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

4.6 Test methods.

4.6.1 Examination of product. Each indicator shall be inspected to determined conformance with this and the applicable detail specification with respect to all requirements not covered by tests.

4.6.2 Leakage.

4.6.2.1 Case leak. The pitot and static pressure connections of the indicator shall be joined with a "Y" connection and shall be connected together to the same mercury manometer and sources of suction and pressure. A suction of 25 inches of mercury shall be applied to the pitot and static pressure connections of the indicator. With the source disconnected for a period of 1 minute, the difference between the mercury levels in the manometer shall not change by more than 0.01 inch of mercury. This procedure shall be repeated with a pressure of 10 inches of mercury. During a period of 1 minute, the difference in the mercury levels in the manometer shall not change by more than 0.01 inch of mercury.

4.6.2.2 Diaphragm capsule leak. With the static connection open to the atmosphere, a pressure sufficient to produce approximately full scale deflection of the pointer, shall be applied to the pitot connection of the indicator at which point the connecting tubing shall be pinched off or otherwise completely sealed. During a period of 1 minute, the pointer shall not change its position by more than 1 mile (statute or nautical, as applicable) per hour.

4.6.3 Scale error at room temperature. (see 6.4.1). The indicator shall be tested for scale errors at the points of the scale indicated by asterisks in the detail specification. The test shall be made by subjecting the indicator to the pressures specified to produce these readings, first with pressure increasing, then with pressures decreasing. With pressures increasing, the pressure shall be brought up to, but shall not exceed the pressures specified to give the desired reading, and with pressures decreasing the pressure shall be brought down to, but shall not fall below the pressures specified to give the desired reading. The errors at the test points shall not exceed those specified for scale error at room temperature in the detail specification. When desired, the indicator shall be tested for scale error at any additional point or points on the scale. The errors at these additional test points shall not exceed those specificed in the detail specification.

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4.6.3.1 Low temperature operation. The indicator shall be subjected to a room temperature scale error test except that during and for 3 hours prior to the test the temperature shall be -35°C. The errors at the test points shall not exceed the tolerance specified by the detail specification.

4.6.3.2 High temperature operation. This test shall be in accordance with 4.6.3.1 except the temperature shall be + 45°C.

4.6.4 Friction error. The indicator shall be tested for friction at every other one of the indicated test points as specified in the detail specification, beginning with the lowest indicated test point. The pressure shall be increased to bring the pointer approximately to the desired readings and then held constant while two readings are taken, the first before the indicator is tapped, the second after. The difference between any two such readings shall not exceed the tolerances specified in the detail specification. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the indicator. This test may be combined with the test for scale error at room temperature.

4.6.5 Position error. The indicator shall be tested as specified in the friction error test for position error at the identical points at which the friction error readings were taken. The error at each position shall not exceed the tolerance specified in the detail specification.

4.6.6 Damping. Sufficient pressure shall be applied to the pitot connection to produce full scale deflection of the airspeed pointer. The pressure shall suddenly be released. The time required for the airspeed pointer to reach the value specified the detail specification shall be equal to the time and within the tolerance specified herein.

4.6.7 Weight. The indicator shall be weighed completely assembled and without connecting fittings and mounting screws. The weight of the indicator shall not exceed the value specified in the detail specification.

4.6.8 Magnetic effect. The indicator shall be held in various positions with its nearest part 5 inches from the center of a short bar magnetic compass whose compensator magnets have been removed and which shall be horizontal intensity is between 0.17 and 0.19 oersted. The various positions shall be so chosen as to include that position which causes the maximum deflection of the compass. The maximum defection shall not exceed 1 degree.

4.6.9 Pointer zero position error. The zero position of the pointer with no differential pressure applied to the indicator shall be within the tolerances specified in the detail specification.

4.6.10 Low temperature exposure. The indicator shall be tested in accordance with Procedure I, Method 502-1 of MIL-STD-810, except that after the 24-hour period and while still at the low temperature, the indicator shall be tested for scale error as specified in the scale error test. The errors at the test points shall not exceed those specified for scale error at room temperature as specified in the detail specification by more than the tolerances specified in the detail specifications. After the indicator has returned to room temperature, it shall pass the individual tests (4.4.1).

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4.6.11 High temperature exposure. The indicator shall be tested in accordance with Procedure I, Method 501.1 of MIL-STD-810. After the indicator has returned to room temperature, it shall pass the individual tests.

4.6.12 Vibration. The indicator shall be tested in accordance with part 1 or part 2, as applicable, of Procedure I, equipment category b.1, Method 514.2 of MIL-STD-810. This test shall be conducted while the differential pressures specified in the detail specification are applied to the indicator. While the indicator is being vibrated, the maximum total spread of pointer vibration and the pointer variation from its original position shall not exceed the tolerances specified in the detail specification.

4.6.13 Overpressure.

4.6.13.1 Overpressure pitot. The pressure specified in the detail specification shall be applied to the pitot pressure connection of the indicator for a period of 10 minutes.

4.6.13.2 Overpressure static. The pressure specified in the detail specification shall be applied to the static connection of the indicator for 10 minutes.

4.6.13.3 Overpressure test. Not less than 5 minutes following the application of the overpressure, the indicator shall be tested for scale error as specified in the scale error at room temperature test (4.6.3). The errors at the test points shall not exceed those specified for scale error at room temperature, as specified in the detail specification by more than the tolerances specified in the detail specification.

4.6.14 Acceleration. The indicator shall be mounted in its normal operating position on a rotating arm, with the rear of the case toward the center of rotation, and shall be subjected to an acceleration of 20g's for a period of 30 seconds. This test shall be repeated with the indicator mounted successively in the following two positions: (1) rotated 90 degrees in azimuth with respect to the original position and (2) dial horizontal, with the zero graduation toward the center of rotation. At the end of this acceleration, the indicator shall pass the individual tests. No damage to the indicator shall result from this test.

4.6.15 Humidity. The indictor shall be mounted in its normal operating position in a chamber and maintained at a temperature of 65°C (149°F) and a minimum relative humidity of 95 percent for 6 hours. External connections shall be made to the indicator to simulate installed conditions. After this period, the heat shall be shut off and the indicator shall be allowed to cool for 18 hours in this atmosphere, in which the humidity rises to 100 percent as the temperature decreases. Precaution shall be taken to prevent dripping on the indicator. The complete procedure shall be repeated 10 times. There shall be no evidence of corrosion or rusting of the internal or external parts of the indicator which would affect the performance of the instrument.

4.6.16 Salt fog. The indicator shall be tested in accordance with Procedure I, Method 509.1 of MIL-STD-810 for 50 hours. External connections shall be made to the indicator to simulate installed conditions. At the end of the 50 hours period, no damage to any part of the indicator which would affect the performance of the instrument shall result from this test. The indicator shall then be subjected to and shall pass the individual tests.

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4.6.17 Seasoning. The indicator shall be subjected to 1,000 applications of a differential pressure sufficient to produce approximately full scale deflection. These application shall be made at a rate not greater than six per minute. Not less than 48 hours following this test, the indicator shall be tested for scale errors as specified in the scale error at room temperature test (4.6.3). The errors at the test points shall not exceed those specified for scale error at room temperature as specified in the detail specification by more than tolerances specified in the detail specification.

4.6.18 Fungus. The indicator shall be tested in accordance with Procedure I, Method 508.1 of MIL-STD-810, except that the test period shall be 14 days. At the option of the contracting activity, tests of component parts of the item may be accepted in lieu of, or in addition to, test, the indicator shall be subjected to and shall pass the aging test (4.6.19). There shall be no deterioration which would adversely affect subsequent operation. This test may be run simultaneously with the aging test.

4.6.19 Aging. The indicator shall be retained for a period of not less than 30 days after having been subjected to the seasoning test, and then tested for scale errors in this test shall not differ from those obtained 48 hours after the 1,000 applications of differential pressure specified in the seasoning test by more than the tolerances specified in the detail specification.

4.6.20 Mounting nuts. Each attached mounting nut, if furnished, of the completely assembled indicator case shall have an axial load of 25 pounds applied so as to tend to pull the nut out of the lug toward the rear of the case. There shall be no loosening of the nut. A torque of 10 pound-inches shall then be applied and the rotation of the nut shall be limited by contact between a flat of the insert nut and the case.

4.6.21 Mounting lugs. The indicator case with the mechansim and cover glass removed shall be mounted in a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug or the mounting nut and attached to a pull strap in the machine. A load of 175 pounds shall be applied to each lug for a period of 1 minute along the mounting hole axis and toward the front of the case. The lugs shall withstand the applied load without fracture.

4.7 Inspection of packaging. Tests of methods of packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 or MIL-P-116.

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5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The airspeed indicators are intended for use on aircraft to indicate the speed relative to the air at sea level, and, when at any other altitude, the equivalent sea level speed corresponding to the actual sustaining force of the air.

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6.2 Ordering data. Procurement documens should specify:

(a) Title, number, and date of the detail specification.

(b) MS part number and quantity desired.

(c) Data requirements, installation instructions (see 3.2, 3.13.1, and 4.3).

(d) Spring nuts, if required (see 3.13.3).

(e) Whether sampling plan B is to be omitted and, if not, the name and address of the laboratory to which the samples should be submitted (see 4.4.2.2.1).

(f) Levels of packaging (see 5.1) and packing (see 5.2) desired.

6.3 Drawing requirements. The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with DOD-D-1000.

6.4 Definitions.

6.4.1 Scale error. "Scale error" is defined as the difference between the indication of the instrument and the calibrated airspeed corresponding to the applied differential pressure as specified in U.S. Standard Atmosphere, 1962.

6.5 External connection. When external connections to sumulate installed conditions are specified, they shall consist of 10 feet of 0.250 copper or aluminum tubing so coiled that moisture can drain out the open end. Similar coils shall be connected to the pitot and static pressure connections.

6.6 Subject term (key word) listing.

Airspeed
Indicator
Pitot-Static

6.7 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement STANAG 3636 and ASCC Air Standard 10/41 and 10/46. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the contracting activity will take appropriate action through internation standardization channels including departmental standardization offices to change the agreement or make other appropriate accomodations.

6.8 Changes from previous issue. Asterisk (or vetical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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AIR FORCE -99
NAVY -AS
ARMY -AV

Preparing activity:
AIR FORCE -71

Project number:
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MIL-I-5415C	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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